

Chapter 2: Review of Probability and Statistics By-Hand Exercises (with Solutions)

1. Problem

Random variables and their realizations

An online shop records the following variables for each of its customers:

- (i) time spent on the website (in minutes),
 - (ii) number of items in the shopping basket,
 - (iii) payment method $\in \{\text{credit card}, \text{PayPal}, \text{invoice}\}$,
 - (iv) customer's body height (in cm),
 - (v) whether the customer returns at least one item $\in \{\text{yes}, \text{no}\}$.
- (a) Decide for each variable whether it is a discrete or a continuous random variable.
 - (b) For which of the variables does it make sense to compute a mean?
 - (c) Which tool describes the distribution of variable (i): a density function (pdf) or a probability mass function (pmf)? Which tool describes the distribution of variable (ii)? Which tool can be used for *both* (i) and (ii)?
 - (d) Let T_i denote the time that customer i spends on the website. Explain the difference between T_i and $T_{i,\text{obs}}$. Which of the two is a fixed number?

Solution

- (a) Continuous: (i) time and (iv) body height; both can take any value in an interval. Discrete: (ii) number of items ($\{0, 1, 2, \dots\}$), (iii) payment method (three possible values) and (v) returns (two possible values).
- (b) Only for numeric variables: (i), (ii) and (iv). The values of (iii) and (v) are categories, and the “average payment method” has no meaning. (If we code (v) as $1 = \text{yes}$ and $0 = \text{no}$, the mean of the 0/1 variable is the share of customers who return at least one item.)
- (c) Variable (i) is continuous, so we use a density $f_T(t)$. Variable (ii) is discrete, so we use a pmf. The cumulative distribution function $F(x) = P(X \leq x)$ works for both, since it is defined for every numeric random variable, whether continuous or discrete.
- (d) T_i is a *random variable*: before we observe customer i , the time spent is uncertain and could take many different values. $T_{i,\text{obs}}$ is the *realization*, i.e., the value we actually observed (for instance $T_{i,\text{obs}} = 4.3$ minutes). Only $T_{i,\text{obs}}$ is a fixed number.

2. Problem

Probability mass function, mean and variance

Let W denote the number of customer complaints that a small company receives on a randomly chosen day. W has the following probability mass function:

w_k	0	1	2	3	4
$p_W(w_k)$	0.10	0.25	0.30	c	0.15

- (a) Determine c such that p_W is a valid probability mass function.
- (b) Compute $P(W \geq 2)$ and $P(W \leq 1)$.
- (c) Compute the expected number of complaints per day, $E(W)$.
- (d) Compute the variance $\text{Var}(W)$ and the standard deviation $\text{SD}(W)$.

- (e) Each complaint costs the company 20 EUR. In addition, the complaint hotline has fixed costs of 150 EUR per day. Thus the daily costs are $K = 150 + 20W$. Compute $E(K)$ and $SD(K)$.

Solution

- (a) The probabilities must add up to one: $c = 1 - (\text{sum of the other four probabilities}) = 0.20$.
- (b) $P(W \geq 2) = p_W(2) + p_W(3) + p_W(4) = 0.65$ and $P(W \leq 1) = p_W(0) + p_W(1) = 0.35$.
Check: $P(W \leq 1) = 1 - P(W \geq 2)$.

(c)

$$E(W) = \sum_k w_k p_W(w_k) = 0 \cdot 0.10 + 1 \cdot 0.25 + 2 \cdot 0.30 + 3 \cdot 0.20 + 4 \cdot 0.15 = 2.05.$$

(d)

$$\begin{aligned} \text{Var}(W) &= \sum_k (w_k - E(W))^2 p_W(w_k) \\ &= (0 - 2.05)^2 \cdot 0.10 + (1 - 2.05)^2 \cdot 0.25 + (2 - 2.05)^2 \cdot 0.30 \\ &\quad + (3 - 2.05)^2 \cdot 0.20 + (4 - 2.05)^2 \cdot 0.15 \\ &= 1.4475, \end{aligned}$$

$$\text{and } SD(W) = \sqrt{1.4475} \approx 1.203.$$

- (e) Using $E(aW + b) = aE(W) + b$ and $SD(aW + b) = |a|SD(W)$:

$$E(K) = 150 + 20 \cdot 2.05 = 191.00 \text{ EUR}, \quad SD(K) = 20 \cdot 1.203 \approx 24.06 \text{ EUR}.$$

The fixed costs shift the mean, but do not affect the spread.

3. Problem

Density function

The delay X (in hours) of a parcel delivery is a continuous random variable with density function

$$f_X(x) = \begin{cases} c \cdot x & \text{for } 0 \leq x \leq 2 \\ 0 & \text{otherwise,} \end{cases}$$

where $c > 0$ is a constant.

- (a) Determine c such that f_X is a valid density function.
- (b) Compute the probability that the delay is between 30 minutes and 90 minutes, i.e., $P(0.5 \leq X \leq 1.5)$.
- (c) Derive the cumulative distribution function $F_X(x) = P(X \leq x)$ for all $x \in \mathbb{R}$. Use it to compute $P(X > 1)$.
- (d) Compute the expected delay $E(X)$.
- (e) Compute $\text{Var}(X)$ and $SD(X)$.
Hint: It is often easier to use $\text{Var}(X) = E(X^2) - (E(X))^2$, where $E(X^2) = \int_{-\infty}^{\infty} x^2 f_X(x) dx$.
- (f) Another delay V (in hours) has the density $f_V(v) = 4$ for $0 \leq v \leq 0.25$ and $f_V(v) = 0$ otherwise. Is this a valid density, even though $f_V(v) > 1$? Explain.

Solution

(a) A density must integrate to one:

$$1 = \int_0^2 c x dx = c \left[\frac{x^2}{2} \right]_0^2 = c \cdot 2 \Rightarrow c = \frac{1}{2}.$$

Since $c > 0$, also $f_X(x) \geq 0$ for all x .

(b)

$$P(0.5 \leq X \leq 1.5) = \int_{0.5}^{1.5} \frac{x}{2} dx = \left[\frac{x^2}{4} \right]_{0.5}^{1.5} = \frac{2.25}{4} - \frac{0.25}{4} = 0.5.$$

(c) For $0 \leq x \leq 2$: $F_X(x) = \int_0^x \frac{t}{2} dt = \frac{x^2}{4}$. Thus

$$F_X(x) = \begin{cases} 0 & x < 0 \\ x^2/4 & 0 \leq x \leq 2 \\ 1 & x > 2. \end{cases}$$

$$P(X > 1) = 1 - P(X \leq 1) = 1 - F_X(1) = 1 - \frac{1}{4} = 0.75.$$

(d)

$$E(X) = \int_0^2 x \cdot \frac{x}{2} dx = \left[\frac{x^3}{6} \right]_0^2 = \frac{8}{6} = \frac{4}{3} \approx 1.33 \text{ hours.}$$

(e) $E(X^2) = \int_0^2 x^2 \cdot \frac{x}{2} dx = \left[\frac{x^4}{8} \right]_0^2 = 2$, so

$$\text{Var}(X) = 2 - \left(\frac{4}{3} \right)^2 = 2 - \frac{16}{9} = \frac{2}{9} \approx 0.222 \quad \text{and} \quad \text{SD}(X) = \sqrt{2/9} \approx 0.471 \text{ hours.}$$

(f) Yes. $f_V(v) \geq 0$ for all v , and $\int_0^{0.25} 4 dv = 4 \cdot 0.25 = 1$. A density value is *not* a probability; only areas under the density are probabilities. Therefore, density values larger than one are no problem.

4. Problem

Empirical cumulative distribution function

A mail-order company records the delivery times (in days) of $n = 8$ randomly selected orders:

$$X_{1,\text{obs}}, \dots, X_{8,\text{obs}} = 7, 1, 5, 5, 2, 2, 2, 5.$$

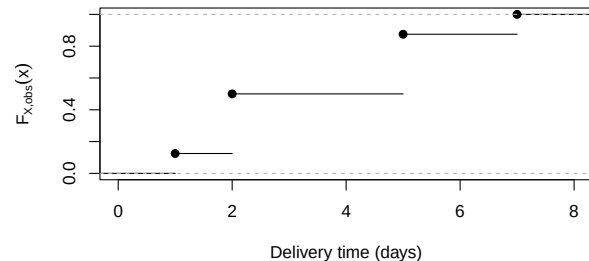
- Compute the empirical cdf $\hat{F}_{X,\text{obs}}(x) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}(X_{i,\text{obs}} \leq x)$ at the points $x = 0, 2, 3.5, 6, 10$.
- Sketch the empirical cdf $\hat{F}_{X,\text{obs}}(x)$ for $x \in [0, 8]$. Where does it jump, and how large are the jumps?
- Use the empirical cdf to estimate the probability that a delivery takes more than 3 days.
- Estimate the probability mass function $p_X(k)$, $k = 1, \dots, 7$, by relative frequencies.

Solution

(a) Count the observations that are smaller than or equal to x and divide by $n = 8$:

x	0	2	3.5	6	10
number of $X_{i,\text{obs}} \leq x$	0	4	4	7	8
$\hat{F}_{X,\text{obs}}(x)$	0.000	0.500	0.500	0.875	1.000

- (b) $\hat{F}_{X,\text{obs}}$ is a step function that starts at 0 and ends at 1. It jumps at each observed value; the jump height at a value k equals (number of observations equal to k)/ n . Between



the observed values it is constant.

- (c) $\hat{P}_{\text{obs}}(X > 3) = 1 - \hat{F}_{X,\text{obs}}(3) = 0.500$.
 (d) The relative frequencies are:

k	1	2	3	4	5	6	7
$\hat{p}_{X,\text{obs}}(k)$	0.125	0.375	0.000	0.000	0.375	0.000	0.125

These are exactly the jump heights of the empirical cdf.

5. Problem

Calculation rules for mean and variance

The monthly sales X (in units) of a product are random with $E(X) = 600$ and $\text{SD}(X) = 40$. Each unit is sold for 15 EUR, and the fixed costs are 3000 EUR per month. The monthly profit is thus

$$\Pi = 15X - 3000.$$

- (a) Compute the expected monthly profit $E(\Pi)$.
 (b) Compute $\text{Var}(\Pi)$ and $\text{SD}(\Pi)$. Why do the fixed costs not matter here?
 (c) The company sells a second product with monthly sales Y and $E(Y) = 250$. What is the expected total number of units sold, $E(X + Y)$? Do you need to assume that X and Y are independent?
 (d) The daily maximum temperature C (in degrees Celsius) in July has mean $E(C) = 20$ and standard deviation $\text{SD}(C) = 3$. In degrees Fahrenheit, the temperature is $F = 1.8C + 32$. Compute $E(F)$ and $\text{SD}(F)$.
 (e) Let X be any random variable with mean μ_X and standard deviation $\sigma_X > 0$. Show that the standardized random variable $Z = \frac{X - \mu_X}{\sigma_X}$ has $E(Z) = 0$ and $\text{Var}(Z) = 1$.

Solution

- (a) $E(\Pi) = 15 \cdot E(X) - 3000 = 15 \cdot 600 - 3000 = 6000$ EUR.
 (b) $\text{Var}(\Pi) = 15^2 \cdot \text{Var}(X) = 225 \cdot 1600 = 360,000$ EUR² and $\text{SD}(\Pi) = 15 \cdot 40 = 600$ EUR.
 The fixed costs only shift the profit by a constant. A constant shift moves the whole distribution but does not change how much the values spread around the mean: $\text{Var}(aX + b) = a^2 \text{Var}(X)$.
 (c) $E(X + Y) = E(X) + E(Y) = 600 + 250 = 850$ units. This rule holds for *any* two random variables; independence is not needed.
 (d) $E(F) = 1.8 \cdot 20 + 32 = 68$ and $\text{SD}(F) = 1.8 \cdot 3 = 5.4$ degrees Fahrenheit.

(e) Write $Z = aX + b$ with $a = 1/\sigma_X$ and $b = -\mu_X/\sigma_X$. Then

$$E(Z) = \frac{1}{\sigma_X} E(X) - \frac{\mu_X}{\sigma_X} = \frac{\mu_X}{\sigma_X} - \frac{\mu_X}{\sigma_X} = 0 \quad \text{and} \quad \text{Var}(Z) = \frac{1}{\sigma_X^2} \text{Var}(X) = \frac{\sigma_X^2}{\sigma_X^2} = 1.$$

6. Problem

Sample mean and sample variance

A bakery records its daily revenue (in 1000 EUR) on $n = 6$ randomly selected days:

$$X_{1,\text{obs}}, \dots, X_{6,\text{obs}} = 11, 9, 11, 6, 8, 9.$$

- Compute the sample mean $\bar{X}_{\text{obs}}$ by hand.
- Compute the sample variance $\hat{\sigma}_{X,\text{obs}}^2$ and the sample standard deviation $\hat{\sigma}_{X,\text{obs}}$ by hand.
- The bakery wants to report the revenue in EUR instead of 1000 EUR. How do the sample mean, the sample variance and the sample standard deviation change?
- Is $\bar{X}_{\text{obs}}$ equal to the true mean $E(X)$ of the daily revenue? Explain.

Solution

(a) $\bar{X}_{\text{obs}} = \frac{1}{6}(11 + 9 + 11 + 6 + 8 + 9) = \frac{54}{6} = 9.$

(b) Deviations from the sample mean:

$X_{i,\text{obs}}$	11	9	11	6	8	9
$X_{i,\text{obs}} - \bar{X}_{\text{obs}}$	2	0	2	-3	-1	0
$(X_{i,\text{obs}} - \bar{X}_{\text{obs}})^2$	4	0	4	9	1	0

$$\hat{\sigma}_{X,\text{obs}}^2 = \frac{1}{n-1} \sum_{i=1}^n (X_{i,\text{obs}} - \bar{X}_{\text{obs}})^2 = \frac{18}{5} = 3.60, \quad \hat{\sigma}_{X,\text{obs}} = \sqrt{3.60} \approx 1.90.$$

Note that the deviations $X_{i,\text{obs}} - \bar{X}_{\text{obs}}$ always sum to zero.

- Multiplying all values by $a = 1000$ multiplies the sample mean and the sample standard deviation by 1000, and the sample variance by 1000^2 : $\bar{X}_{\text{obs}} = 9,000$ EUR, $\hat{\sigma}_{X,\text{obs}}^2 \approx 3,600,000$ EUR², $\hat{\sigma}_{X,\text{obs}} \approx 1,897$ EUR. (These are the sample versions of the rules $E(aX + b) = aE(X) + b$, $\text{Var}(aX + b) = a^2 \text{Var}(X)$ and $\text{SD}(aX + b) = |a| \text{SD}(X)$.)
- No (or only by chance). $\bar{X}_{\text{obs}}$ is an *estimate* of the usually unknown $E(X)$, computed from one particular sample of $n = 6$ days. A different sample of days would give a different value (sampling variability). By the Law of Large Numbers, $\bar{X}_{\text{obs}}$ gets close to $E(X)$ when n is large.

7. Problem

Covariance and correlation

A retail chain records the advertising expenditures X (in 1000 EUR) and the sales Y (in 1000 units) of $n = 5$ of its stores:

Store i	1	2	3	4	5
$X_{i,\text{obs}}$	5	6	8	7	4
$Y_{i,\text{obs}}$	17	18	24	20	16

- Compute the sample covariance $\widehat{\text{Cov}}_{\text{obs}}(X, Y)$ by hand.
- Compute the sample correlation $\widehat{\text{Cor}}_{\text{obs}}(X, Y)$ by hand and interpret it.

- (c) Suppose the advertising expenditures are recorded in EUR (instead of 1000 EUR), and the sales in units (instead of 1000 units). How do the covariance and the correlation change?

Hint: $\text{Cov}(a_X X + b_X, a_Y Y + b_Y) = a_X a_Y \text{Cov}(X, Y)$.

- (d) Does the result in (b) show that advertising *causes* higher sales? Explain.

Solution

- (a) Sample means: $\bar{X}_{\text{obs}} = 6$ and $\bar{Y}_{\text{obs}} = 19$.

i	1	2	3	4	5	Sum
$X_{i,\text{obs}} - \bar{X}_{\text{obs}}$	-1	0	2	1	-2	0
$Y_{i,\text{obs}} - \bar{Y}_{\text{obs}}$	-2	-1	5	1	-3	0
$(X_{i,\text{obs}} - \bar{X}_{\text{obs}})(Y_{i,\text{obs}} - \bar{Y}_{\text{obs}})$	2	0	10	1	6	19
$(X_{i,\text{obs}} - \bar{X}_{\text{obs}})^2$	1	0	4	1	4	10
$(Y_{i,\text{obs}} - \bar{Y}_{\text{obs}})^2$	4	1	25	1	9	40

$$\widehat{\text{Cov}}_{\text{obs}}(X, Y) = \frac{1}{n-1} \sum_{i=1}^n (X_{i,\text{obs}} - \bar{X}_{\text{obs}})(Y_{i,\text{obs}} - \bar{Y}_{\text{obs}}) = \frac{19}{4} = 4.75.$$

- (b) With $\hat{\sigma}_{X,\text{obs}}^2 = 10/4 = 2.50$ and $\hat{\sigma}_{Y,\text{obs}}^2 = 40/4 = 10.00$:

$$\widehat{\text{Cor}}_{\text{obs}}(X, Y) = \frac{4.75}{\sqrt{2.50 \cdot 10.00}} \approx 0.950.$$

The correlation is positive and strong: stores with higher advertising expenditures tend to have higher sales. The points lie close to a straight line.

- (c) With $a_X = 1000$ and $a_Y = 1000$, the covariance is multiplied by $1000 \cdot 1000 = 10^6$, i.e., it becomes $\approx 4,750,000$. The covariance depends on the units of measurement. The correlation does not change, because the factors cancel:

$$\frac{a_X a_Y \text{Cov}(X, Y)}{\sqrt{a_X^2 \text{Var}(X) a_Y^2 \text{Var}(Y)}} = \text{Cor}(X, Y) \quad \text{for } a_X, a_Y > 0.$$

This is why the correlation, and not the covariance, is used to judge the strength of a linear association.

- (d) No. A correlation only describes a (linear) statistical association. For instance, the chain may spend more money on advertising in large stores in busy city centers, which also have more sales for other reasons (a confounder: store size or location). Moreover, $n = 5$ stores is a very small sample, so the estimate is quite uncertain.

8. Problem

Joint, marginal and conditional distributions

An online shop classifies its customers by $X \in \{0, 1\}$, where $X = 0$ denotes a new customer and $X = 1$ a returning customer. Let $Y \in \{1, 2, 3\}$ denote the number of items a customer buys. The joint probability mass function $p_{Y,X}(y, x) = P(Y = y, X = x)$ is given by

	$Y = 1$	$Y = 2$	$Y = 3$
$X = 0$	0.20	0.15	0.05
$X = 1$	0.10	0.25	0.25

For discrete random variables, all formulas for densities carry over with sums instead of integrals. For instance, the conditional pmf is $p_{Y|X}(y | x) = p_{Y,X}(y, x)/p_X(x)$.

- (a) Check that $p_{Y,X}$ is a valid joint probability mass function.

- (b) Compute the marginal pmfs $p_X(x)$ and $p_Y(y)$.
- (c) Compute the conditional pmf of Y given $X = 0$, and the conditional pmf of Y given $X = 1$.
- (d) Compute the conditional means $E(Y | X = 0)$ and $E(Y | X = 1)$, as well as the unconditional mean $E(Y)$. Interpret the results.
- (e) Are Y and X independent? Justify your answer.

Solution

- (a) All entries are non-negative, and $0.20 + 0.15 + 0.05 + 0.10 + 0.25 + 0.25 = 1$.
- (b) Marginal pmfs are row and column sums:

$$p_X(0) = 0.20 + 0.15 + 0.05 = 0.40, \quad p_X(1) = 0.10 + 0.25 + 0.25 = 0.60,$$

$$p_Y(1) = 0.30, \quad p_Y(2) = 0.40, \quad p_Y(3) = 0.30.$$

- (c) Divide each row by its row sum:

	$y = 1$	$y = 2$	$y = 3$
$p_{Y X}(y 0)$	$0.20/0.40 = 0.500$	$0.15/0.40 = 0.375$	$0.05/0.40 = 0.125$
$p_{Y X}(y 1)$	$0.10/0.60 \approx 0.167$	$0.25/0.60 \approx 0.417$	$0.25/0.60 \approx 0.417$

Each row sums to one; a conditional pmf is itself a proper pmf.

- (d)

$$E(Y | X = 0) = 1 \cdot 0.500 + 2 \cdot 0.375 + 3 \cdot 0.125 = 1.625,$$

$$E(Y | X = 1) = 1 \cdot \frac{1}{6} + 2 \cdot \frac{5}{12} + 3 \cdot \frac{5}{12} = \frac{27}{12} = 2.25,$$

$$E(Y) = 1 \cdot 0.30 + 2 \cdot 0.40 + 3 \cdot 0.30 = 2.$$

On average, new customers buy 1.625 items and returning customers 2.25 items. The conditional mean $E(Y | X = x)$ as a function of x is the regression function: it tells us how the average number of items changes with the customer type.

- (e) No. For independence we would need $p_{Y,X}(y, x) = p_Y(y) p_X(x)$ for *all* (y, x) . But, e.g., $p_{Y,X}(1, 0) = 0.20 \neq p_Y(1) p_X(0) = 0.30 \cdot 0.40 = 0.12$. Equivalently: the conditional pmfs in (c) differ from each other (and from p_Y), so knowing X changes the distribution of Y .

9. Problem**The normal distribution**

A machine fills coffee into cups. The filled amount X (in ml) is normally distributed, $X \sim \mathcal{N}(\mu, \sigma^2)$ with $\mu = 200$ and $\sigma = 5$. Let $\Phi(z) = P(Z \leq z)$ denote the cdf of the standard normal distribution $Z \sim \mathcal{N}(0, 1)$. Use a table of Φ and the following quantiles: $\Phi(-1.645) = 0.05$ and $\Phi(-2.326) = 0.01$.

- (a) Compute the probability that a cup contains less than 188 ml.
- (b) Compute the probability that a cup contains between 195 ml and 205 ml.
Hint: By symmetry, $\Phi(-z) = 1 - \Phi(z)$.
- (c) Which amount is undercut by only 5% of all cups? That is, find x such that $P(X \leq x) = 0.05$.
- (d) The company wants that at most 1% of all cups contain less than 188 ml. The standard deviation $\sigma = 5$ cannot be changed, but the machine's mean μ can be adjusted. How large must μ be at least?

Solution

Standardizing gives $Z = (X - \mu)/\sigma \sim \mathcal{N}(0, 1)$.

(a)

$$P(X < 188) = P\left(Z < \frac{188 - 200}{5}\right) = \Phi(-2.40) \approx 0.0082.$$

(b)

$$\begin{aligned} P(195 \leq X \leq 205) &= P\left(-\frac{5}{5} \leq Z \leq \frac{5}{5}\right) = \Phi(1.00) - \Phi(-1.00) \\ &= 2\Phi(1.00) - 1 \approx 2 \cdot 0.8413 - 1 = 0.6827. \end{aligned}$$

(c) We need the 5% quantile of X . Since $\Phi(-1.645) = 0.05$,

$$0.05 = P(X \leq x) = \Phi\left(\frac{x - 200}{5}\right) \Rightarrow \frac{x - 200}{5} = -1.645 \Rightarrow x = 200 - 1.645 \cdot 5 \approx 191.78 \text{ ml.}$$

(d) We need $P(X < 188) = \Phi((188 - \mu)/5) \leq 0.01 = \Phi(-2.326)$. Since Φ is increasing, this requires

$$\frac{188 - \mu}{5} \leq -2.326 \Leftrightarrow \mu \geq 188 + 2.326 \cdot 5 \approx 199.63 \text{ ml.}$$

10. Problem**The data-generating process**

The monthly electricity costs Y_i (in EUR) and the living space X_i (in m^2) of households $i = 1, \dots, n$ are generated by the DGP

$$Y_i = \beta_1 + \beta_2 X_i + \varepsilon_i, \quad \varepsilon_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2),$$

with $\beta_1 = 40$, $\beta_2 = 0.8$, $\sigma = 10$, and $X_i \sim \mathcal{N}(80, 20^2)$ independent of ε_i .

- Which variable is the outcome and which is the predictor? What does the error term ε_i capture? Interpret β_2 .
- Compute the values of the regression function $m(x) = E(Y_i | X_i = x)$ at $x = 50$ and $x = 100$.
- Which distribution does Y_i have, given that $X_i = 100$? Compute the probability that a household with 100 m^2 has monthly electricity costs above 140 EUR. (Use a table of the standard normal cdf Φ .)
- In practice, which parts of this DGP are usually unknown?
- The DGP assumes i.i.d. sampling. What does this mean here? Give an example of a data collection where the i.i.d. assumption would be violated.

Solution

- Outcome: electricity costs Y_i . Predictor: living space X_i . The error term ε_i captures everything that affects the electricity costs apart from living space (number of persons, appliances, habits, ...). $\beta_2 = 0.8$: households that differ by 1 m^2 in living space differ, on average, by 0.8 EUR in their monthly electricity costs.
- $m(50) = 40 + 0.8 \cdot 50 = 80$ EUR and $m(100) = 40 + 0.8 \cdot 100 = 120$ EUR.
- Given $X_i = 100$, we have $Y_i = 120 + \varepsilon_i$, thus $Y_i | X_i = 100 \sim \mathcal{N}(120, 10^2)$. Therefore,

$$P(Y_i > 140 | X_i = 100) = 1 - \Phi\left(\frac{140 - 120}{10}\right) = 1 - \Phi(2.00) \approx 0.0228.$$

- (d) Usually unknown are the parameters β_1 , β_2 and σ^2 , the distribution of the predictor X_i , and whether the assumptions of the DGP (linear regression function, normal errors, i.i.d. sampling) are correct at all. We only observe one realization $(Y_{1,\text{obs}}, X_{1,\text{obs}}), \dots, (Y_{n,\text{obs}}, X_{n,\text{obs}})$.
- (e) i.i.d.: (1) The households are *independent*: the living space and electricity costs of one household tell us nothing about those of another household. (2) All pairs (Y_i, X_i) are *identically distributed*, i.e., drawn from the same joint distribution.
Violations: we observe the same household in 12 consecutive months (the observations of the same household are dependent), or we sample several households from the same building with a common heating system; or we sample households from different countries with very different electricity prices (not identically distributed).

11. Problem

Association is not causation

A kiosk next to an outdoor swimming pool records, for each day i , its ice cream sales X_i (in 100 portions) and the number of visitors Y_i of the swimming pool (in 100 persons). Consider the following DGP, where Z_i is the daily maximum temperature (in degrees Celsius):

$$\begin{aligned} Z_i &\sim \mathcal{N}(22, 5^2), \\ X_i &= 2 + 0.5 Z_i + u_i, & u_i &\sim \mathcal{N}(0, 1), \\ Y_i &= 1 + 0.3 Z_i + v_i, & v_i &\sim \mathcal{N}(0, 1), \end{aligned}$$

with Z_i , u_i and v_i independent of each other.

- Does the number of pool visitors Y_i depend on the ice cream sales X_i in this DGP?
- Nevertheless, data from this DGP show a strong positive correlation between X_i and Y_i . Explain why. What is the role of Z_i ?
- Suppose we only use days with (almost) the same maximum temperature, e.g., between 24 and 26 degrees Celsius. What correlation between X_i and Y_i do you expect for these days? Explain.
- Compute $E(X_i)$ and $E(Y_i)$.
- Companies that spend a lot of money on management consultants have, on average, lower profits than other companies. Does this show that consultants are harmful? Give an alternative explanation.

Solution

- No. Y_i only depends on the temperature Z_i and on v_i . Selling more ice cream does not bring more visitors to the pool.
- The temperature Z_i is a **confounder**: it influences both X_i and Y_i . On hot days, more ice cream is sold *and* more people go to the pool; on cold days, both are low. This creates a strong positive association between X_i and Y_i , although there is no causal effect of X_i on Y_i .
- A correlation close to zero. If Z_i is (almost) fixed, the remaining variation in X_i comes from u_i and the remaining variation in Y_i from v_i . Since u_i and v_i are independent, there is (almost) no association left between X_i and Y_i .
- Using $E(aZ + b) = aE(Z) + b$ and $E(X + Y) = E(X) + E(Y)$: $E(X_i) = 2 + 0.5 \cdot 22 + E(u_i) = 2 + 11 + 0 = 13$ and $E(Y_i) = 1 + 0.3 \cdot 22 + E(v_i) = 7.6$.
- No. Companies that are already in economic trouble are more likely to hire consultants. The low profits may thus be the reason for hiring consultants (reverse causality), or both may be driven by a third factor such as a crisis in the company's industry (confounder). The observed association alone cannot tell us the causal effect of consultants on profits.